Optimization with inequality constraints

Delhi School of Economics

Optimization

Readings

- Sundaram: Chapter 6.1–6.4

Optimization with inequality constraints

- The objective function is $f : \mathbb{R}^L \to \mathbb{R}$.
- K constraint functions:

$$h_1 : h : \mathbb{R}^L \to \mathbb{R}, \quad h_2 : \mathbb{R}^L \to \mathbb{R}, \ \ldots, \ h_K : \mathbb{R}^L \to \mathbb{R}.$$

- Problem (standard form):

$$\begin{aligned} &\text{maximize} && f(x) \\ &\text{subject to} && x \in \mathbb{R}^L, \\ & && h_1(x) \geq 0, \ h_2(x) \geq 0, \ \ldots, \ h_K(x) \geq 0. \end{aligned}$$

First order necessary condition: Kuhn–Tucker theorem (statement)

Suppose that $f, h_1, \ldots, h_K$ are C^1 functions. Let x^* be a local maximum of f subject to the inequality constraints. Then there exist multipliers $\lambda_1^*, \ldots, \lambda_K^* \in \mathbb{R}$ such that the following hold:

1. [(KT1)] Complementary slackness and sign condition:

$$\lambda_j^* \geq 0, \qquad \lambda_j^* \, h_j(x^*) = 0, \quad j = 1, \ldots, K.$$

2. [(KT2)] Stationarity:

$$Df(x^*) + \sum_{j=1}^{K} \lambda_j^* \, Dh_j(x^*) = 0.$$

Kuhn–Tucker: remark for minima

If x^* is a local minimum instead, an equivalent form of the conditions can be written with the sign of the multipliers reversed. One common convention is to require $\lambda_j^* \leq 0$ and

$$\lambda_j^* \, h_j(x^*) = 0, \quad j = 1, \ldots, K,$$

together with the same stationarity condition

$$Df(x^*) + \sum_{j=1}^{K} \lambda_j^* \, Dh_j(x^*) = 0.$$

The choice of sign convention depends on how the Lagrangian is formed. Be consistent with your Lagrangian sign when applying the theorem.

Finding a global maximum (practical steps)

1. Use Weierstrass' theorem or another argument to show that a global maximum (or minimum) exists.
2. Find all candidate local maxima using the first order Kuhn–Tucker conditions.
3. Evaluate $f(x)$ at all candidate points obtained in step 2 and compare values to determine the global maximum.

Note: Second order conditions are useful but omitted here.